

# Enrichment analysis

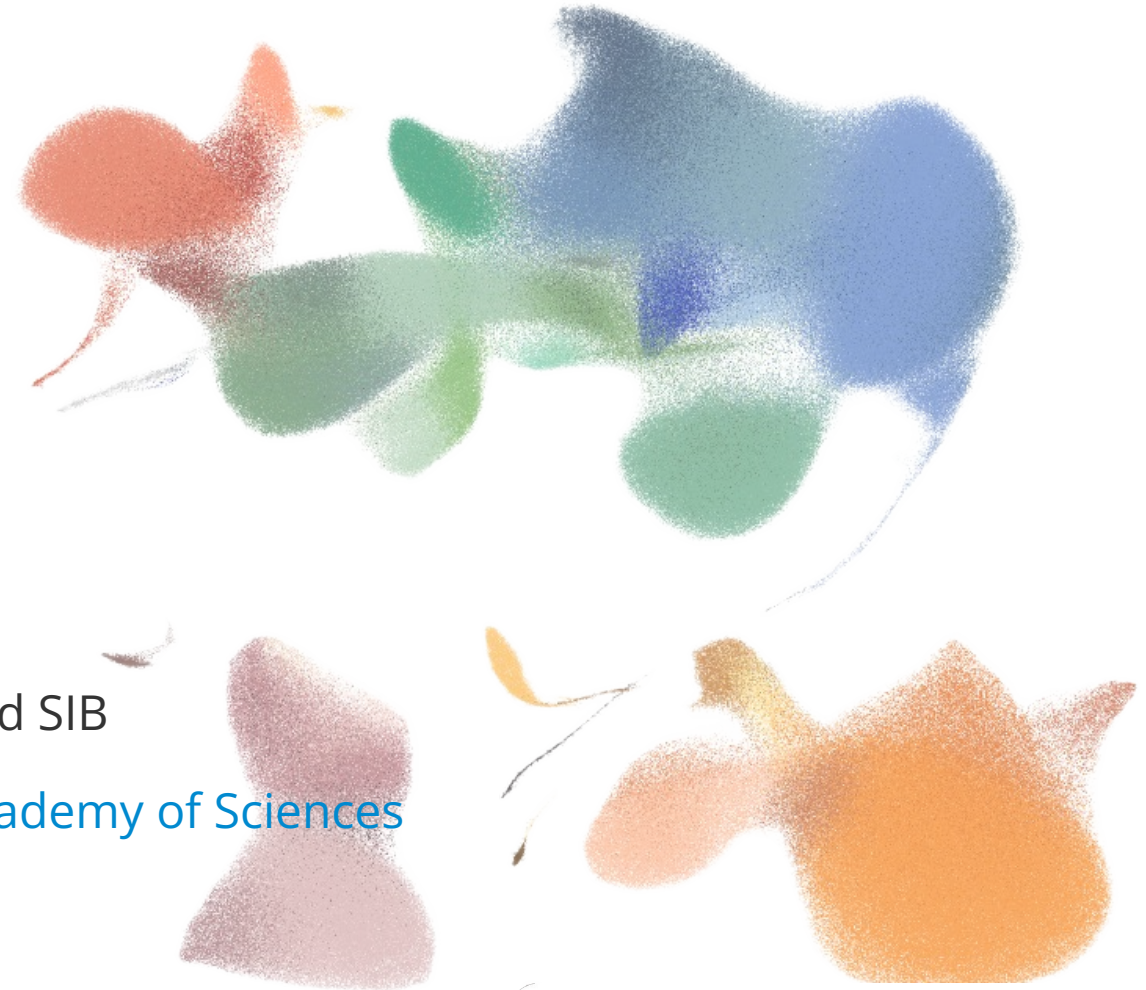
COURSE ON SCRNA-SEQ DATA ANALYSIS

Making sense of DEA results

Jan Kubovčíak · 7.-10. September 2026

with acknowledgments to Luciano Cascione and SIB

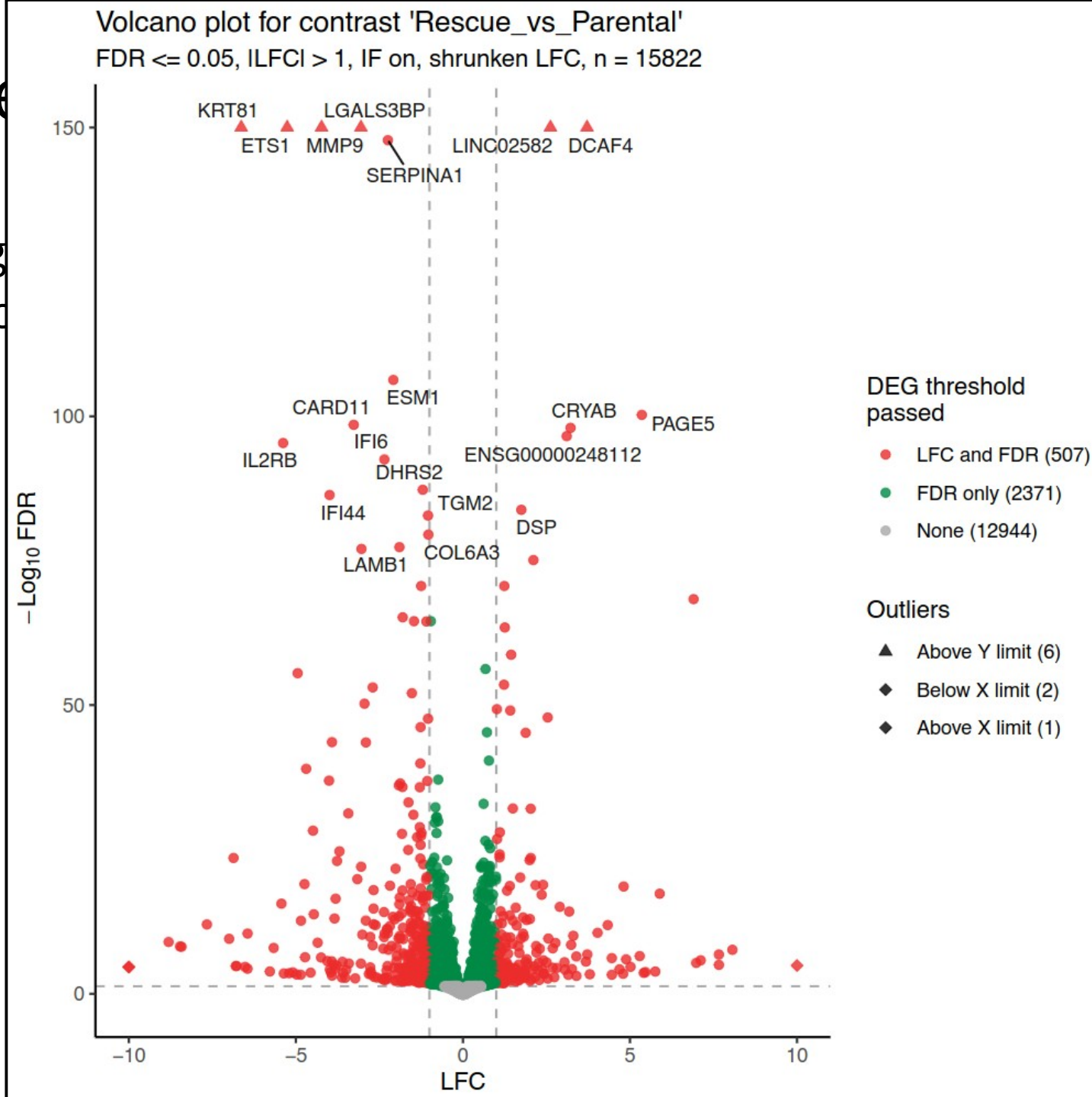
Institute of Molecular Genetics of the Czech Academy of Sciences



# Enrichment

- Find biological groups of

between



# What is a gene set?

- A gene set is an unordered collection of genes that are functionally related.
  - Genes located in the same compartment in a cell (e.g. all proteins located in the cell nucleus)
  - Proteins that are all regulated by a same transcription factor
  - Pathways
  - Custom gene list that comes from a publication and that are down-regulated in a mutant
  - List of genes that contain SNPs associated with a disease
  - ...etc!
- Gene sets are grouped into Knowledge bases

# Gene sets resources

- [Gene Ontology](#)
  - Biological process
  - Molecular function
  - Cellular component
- [KEGG](#)
- [MSigDB](#)
  - Hallmark gene sets
    - Positional gene sets
    - Immunological sets etc.
- [Reactome](#)
- [WikiPathways](#)

# Methods

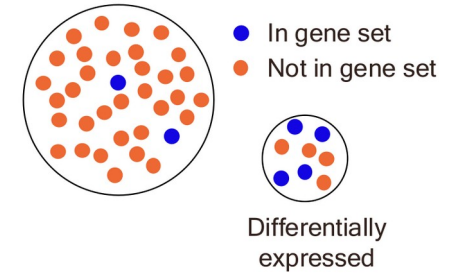
- Over-representation analysis (ORA)
  - Fisher's test of proportions of DEGs and gene set
- Gene set enrichment analysis (classical GSEA)
  - Enrichment score of gene set in ranked list of genes based on deregulation metric

# Over-representation analysis (ORA)

| 2X2<br>count table | Differentially<br>expressed | Not<br>Differentially<br>expressed | total |
|--------------------|-----------------------------|------------------------------------|-------|
| blue               | 2                           | 3                                  | 5     |
| Not blue           | 5                           | 12                                 | 17    |
| total              | 7                           | 15                                 | 22    |

$$2/7 = 0.29 \quad 3/15 = 0.20$$

Fisher's test



$H_0$ : The proportion of genes in the gene set is the same for both groups

$H_a$ : The proportion of genes in the gene set is higher in the differentially expressed group

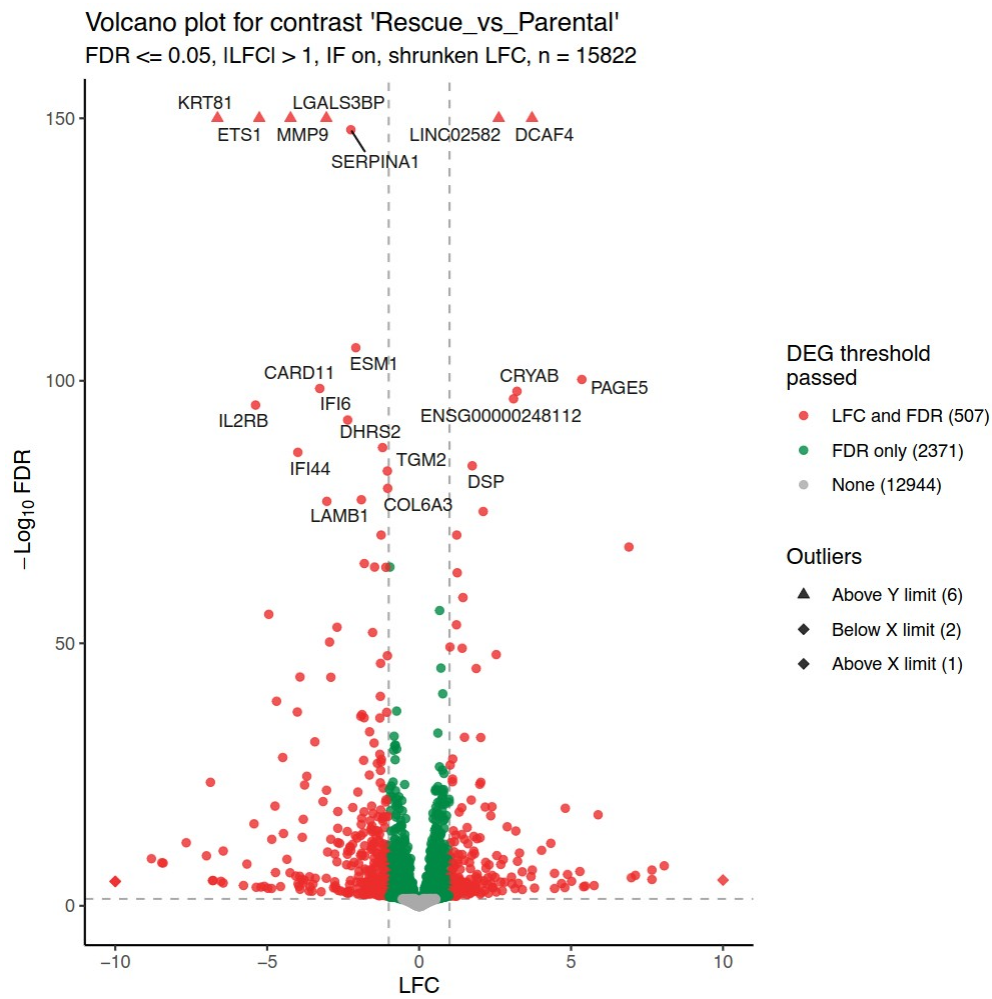
```
> cont.table<-matrix(c(2,3,5,12), ncol=2, byrow = T)  
> fisher.test(cont.table)
```

## Fisher's Exact Test for Count Data

```
data: cont.table  
p-value = 1  
alternative hypothesis: true odds ratio is not equal to 1  
95 percent confidence interval:  
 0.1012333 18.7696686  
sample estimates:  
odds ratio  
 1.56456
```

# ORA

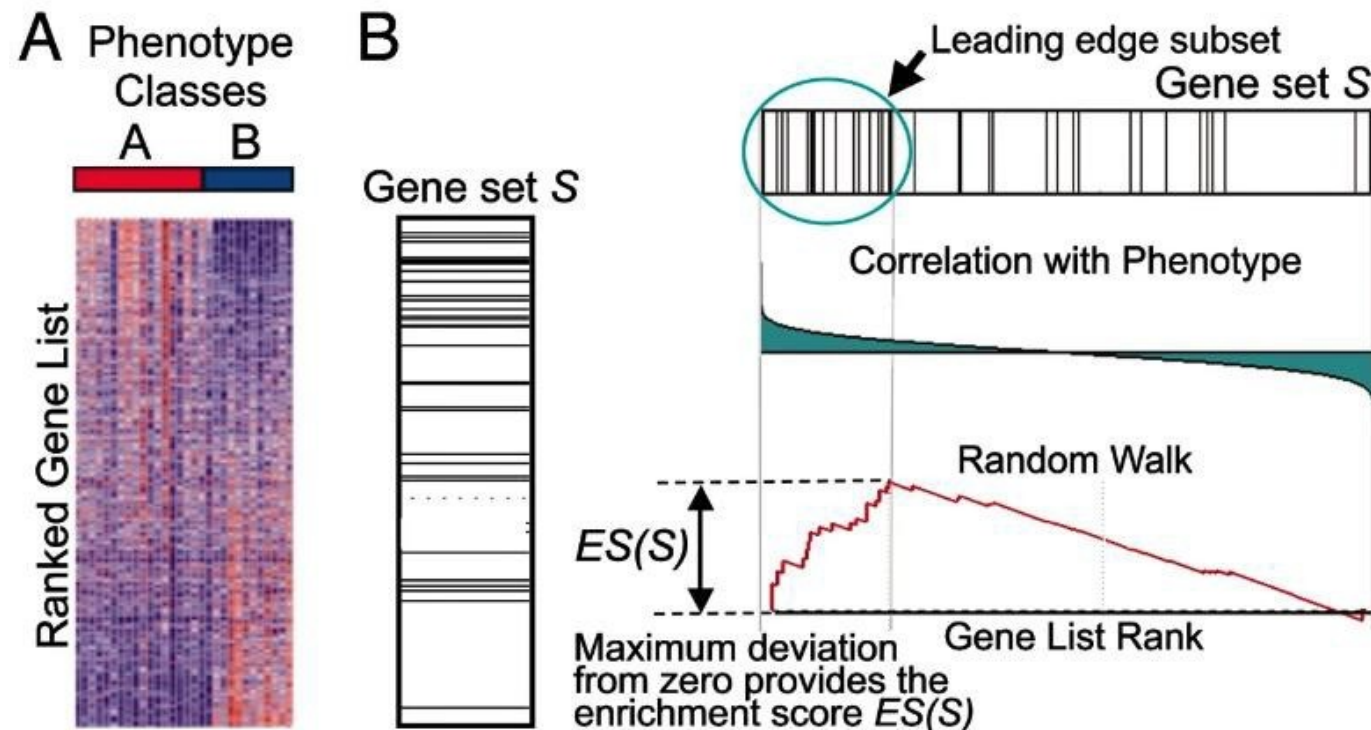
- Multiple gene sets tested = p-value adjustments
- Sensitive to DEG thresholds





# Gene set enrichment analysis (GSEA)

- ORA fails to detect situations where all genes in a predefined set change in a small but coordinated way
- GSEA captures magnitude of deregulation by ranking genes by phenotype





# How to rank the genes?

Research Article | [Open access](#) | Published: 12 May 2017

## Ranking metrics in gene set enrichment analysis: do they matter?

[Joanna Zyla](#), [Michal Marczyk](#) , [January Weiner](#) & [Joanna Polanska](#)

[BMC Bioinformatics](#) **18**, Article number: 256 (2017) | [Cite this article](#)

**45k** Accesses | **43** Citations | **22** Altmetric | [Metrics](#)

### Abstract

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#### Background

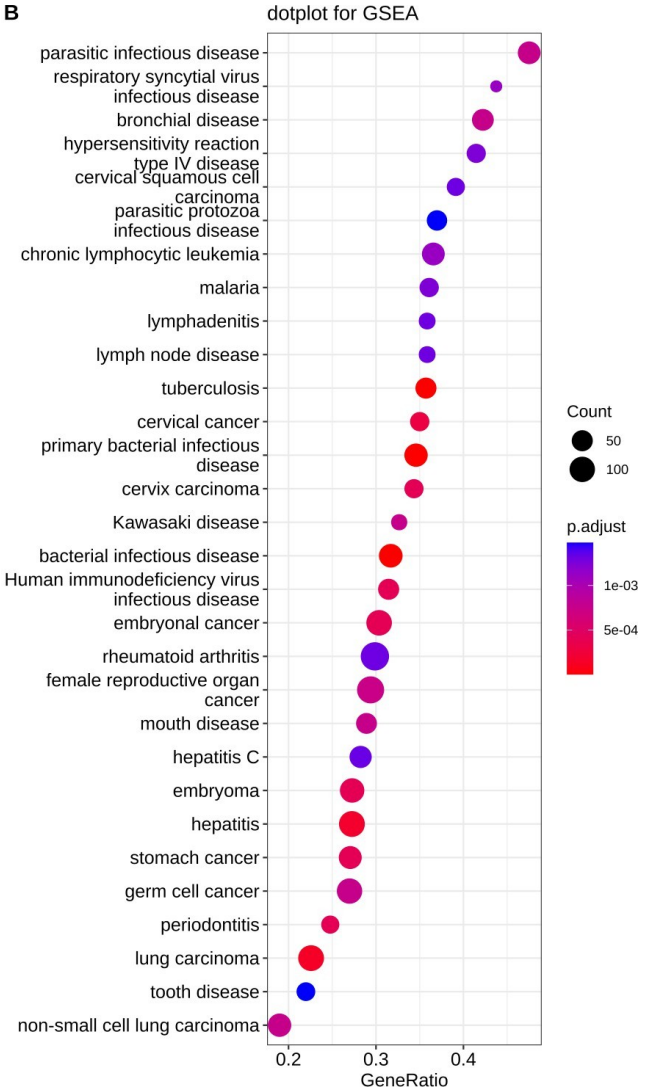
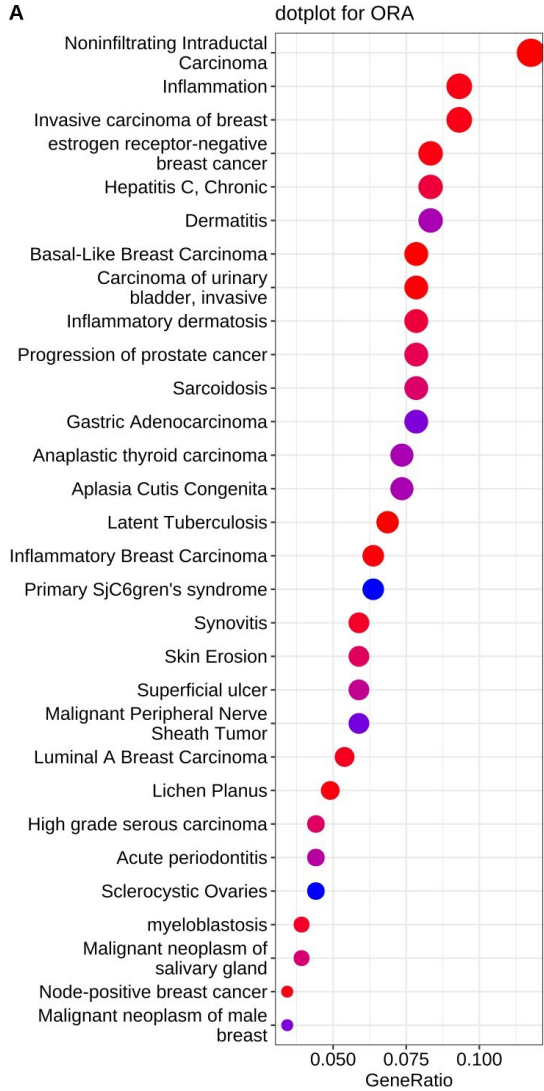
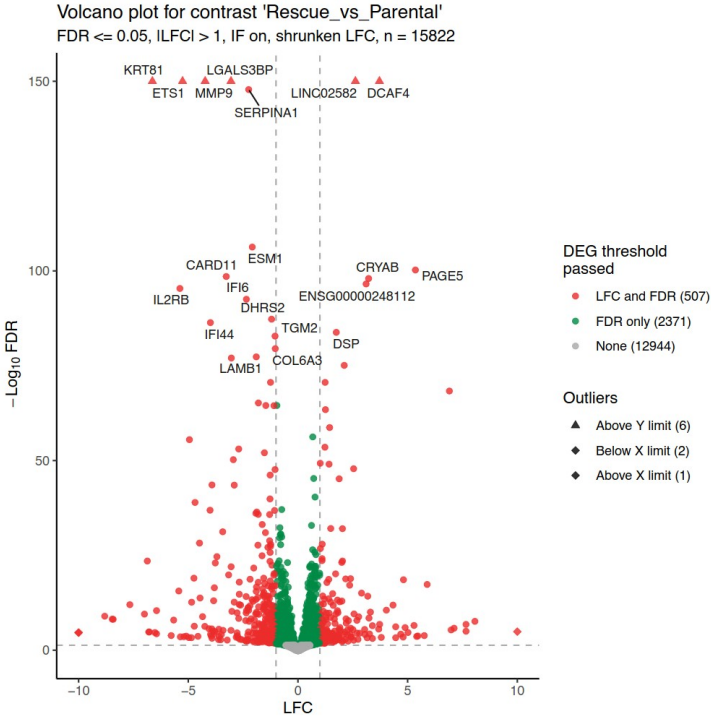
There exist many methods for describing the complex relation between changes of gene expression in molecular pathways or gene ontologies under different experimental conditions. Among them, Gene Set Enrichment Analysis seems to be one of the most commonly used (over 10,000 citations). An important parameter, which could affect the final result, is the choice of a metric for the ranking of genes. Applying a default ranking metric may lead to poor results.

# Implementation of ORA and GSEA

- [clusterProfiler](#)
- Online tools for quick overview (e.g. [Enrichr](#))

|      | KEGG         | GO terms   | custom     |
|------|--------------|------------|------------|
| ORA  | enrichKEGG() | enrichGO() | enricher() |
| GSEA | gseKEGG()    | gseGO()    | GSEA()     |

# Results



Thank you for your attention!